

Partie A:

$$\begin{aligned} 1) \quad -\left(-\frac{9}{5}\right)^4 &= -\left(\frac{9}{5}\right)^4 & ; \quad -5^0 &= -1 & ; \quad (-10)^4 &= 10^4 \\ &= -\left(\frac{9^4}{5^4}\right) & ; \quad \left(\frac{7}{5}\right)^{-1} &= \left(\frac{5}{7}\right)^1 & ; &= 10000 \\ &= -\frac{6561}{625} & ; &= \frac{5}{7} & ; \quad 10^{-3} &= 0,001 \\ & & & & ; \quad \left(\frac{2}{3}\right)^{-3} &= \left(\frac{3}{2}\right)^3 \\ & & & & &= \frac{3^3}{2^3} \\ & & & & &= \frac{27}{8} \end{aligned}$$

$$\begin{aligned} 2) \quad \text{a) On a: } abc &= -1 \\ \text{alors } a \times abc &= (-1) \times a \\ \text{donc } a^2 \cdot bc &= -a \end{aligned}$$

$$\begin{aligned} \text{b) On a } a^2 \cdot b^4 \cdot c^2 &= a^2 \cdot b^2 \cdot c^2 \cdot b^2 \\ &= a^2 \cdot b^2 \cdot c^2 \times b^2 \\ &= (abc)^2 \times b^2 \\ &= (-1)^2 \times b^2 \\ &= 1 \times b^2 \\ &= b^2 \end{aligned}$$

$$\begin{aligned} 3) \quad A &= \frac{(3^5)^4 \times 2^{-5} \times 3^2 \times 4}{2^{17} \times 9} \\ A &= \frac{3^{20} \times 2^{-5} \times \cancel{3^2} \times 2^2}{2^{17} \times \cancel{3^2}} \end{aligned}$$

$$A = 3^{20} \times 2^{-5+2-17}$$

$$A = 3^{20} \times 2^{-20}$$

$$A = \left(\frac{3}{2}\right)^{20}$$

$$; \quad B = \frac{5^2 \times 7^3 \times (5^{-2})^9}{49 \times 5^{-10} \times 35}$$

$$; \quad B = \frac{5^2 \times 7^3 \times 5^{-18}}{7^2 \times 5^{-10} \times 7 \times 5}$$

$$; \quad B = \frac{5^{2-18} \times 7^3}{7^{2+1} \times 5^{-10+1}}$$

$$; \quad B = \frac{5^{-16}}{5^{-9}}$$

$$; \quad B = 5^{-16+9}$$

$$B = 5^{-7}$$

$$C = 4^{11} \times \left(\frac{-2}{3}\right)^6 \times (0,25)^{11} \times \left(\frac{3}{2}\right)^3$$

$$C = (2^2)^{11} \times \left(\frac{-2}{3}\right)^6 \times \left(\frac{1}{4}\right)^{11} \times \left(\frac{3}{2}\right)^3$$

$$C = 2^{22} \times \frac{2^6}{3^6} \times \frac{1}{4^{11}} \times \frac{3^3}{2^3}$$

$$C = \frac{2^{22+6} \times 1 \times 3^3}{3^6 \times 2^{22} \times 2^3}$$

$$C = \frac{2^3 \times 3^3}{3^6 \times 1}$$

$$C = \left(\frac{2}{3}\right)^3$$

Partie B:

$$D = \frac{3^2}{\left(\frac{13}{3}\right)^{-2}} \times [(-13)^{-6}]^{-1}$$

$$D = 3^2 \times \frac{3^2}{13^2} \times \left[\frac{1}{(-13)^6}\right]^{-1}$$

$$D = \frac{3^4}{13^2} \times \left(\frac{1}{13}\right)^{6 \times (-1)}$$

$$D = \frac{3^4}{13^2} \times 13^6$$

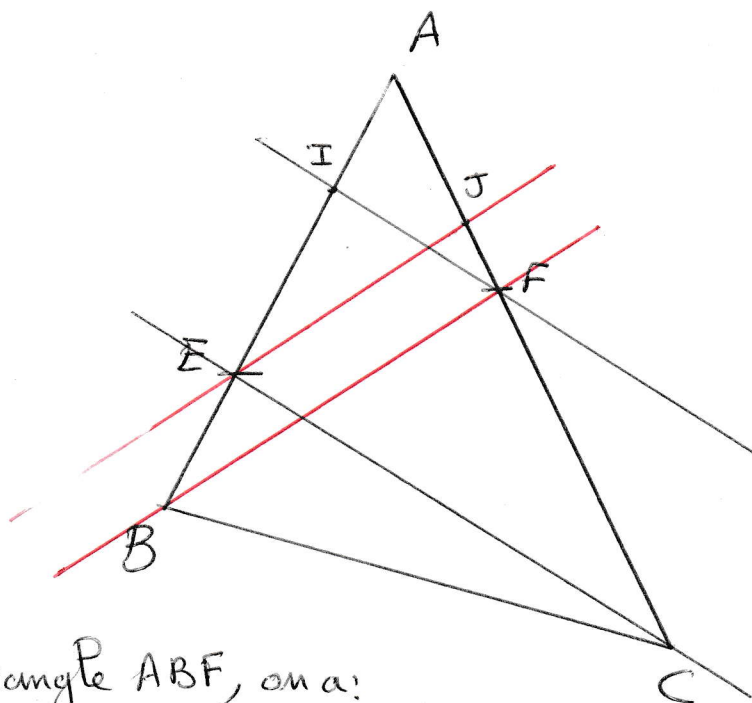
$$D = 3^4 \times 13^4$$

$$D = (3 \times 13)^4$$

$$D = 39^4$$

Partie B:

1)



2) Dans le triangle ABF, on a:

$E \in (AB)$, $J \in (AF)$ et $(EJ) \parallel (BF)$

$$\text{alors } \frac{AE}{AB} = \frac{AJ}{AF} = \frac{EJ}{BF}.$$

$$\text{d'où } \frac{AJ}{AF} = \frac{AE}{AB}.$$

3) Dans le triangle AEC, on a:

$I \in (AE)$, $F \in (AC)$ et $(IF) \parallel (EC)$

$$\text{alors } \frac{AI}{AE} = \frac{AF}{AC} = \frac{IF}{EC}.$$

$$\text{d'où } \frac{AI}{AE} = \frac{AF}{AC}.$$

$$4) \text{ On a } \frac{AJ}{AF} = \frac{AE}{AB}, \text{ alors } AJ \times AB = \underline{AF \times AE}.$$

$$\text{et } \frac{AI}{AE} = \frac{AF}{AC}, \text{ alors } AI \times AC = \underline{AE \times AF}.$$

$$\text{d'où } AE \times AF = AJ \times AB = AI \times AC$$

$$\text{donc } AJ \times AB = AI \times AC.$$